Started on	Wednesday, 9 July 2025, 10:40 AM
State	Finished
Completed on	Wednesday, 9 July 2025, 10:45 AM
Time taken	5 mins 2 secs
Grade	**15.00** out of 15.00 (**100**%)

Question 1

Correct

Mark 3.00 out of 3.00

In solving a kinematic problem with a manipulator, it is found that for a certain joint angle θ, where $-180^\circ \leq \theta \leq 180^\circ$, two function values can be determined:

- $\cos\theta = -0.866025403784439$
- $\sin\theta = 0.5$

Determine θ in degrees, accurately to four significant digits.

Answer: 150

The correct answer is: 150

Question **2**

Correct

Mark 3.00 out of 3.00

Indicate which option regarding the computed torque control technique is false.

- a. The computed torque control technique results in joint torques that can be clearly split into feedforward and a feedback parts.
- b. The computed torque control technique is an example of feedback linearization.
- c. The computed torque control technique uses $\ddot{\theta}_d$ in its control law, where $\theta_d(t)$ is the desired joint angle trajectory. For this reason $\theta_d(t)$ must be at least twice differentiable.
- d. The use of the computed torque control technique is supported by good experimental results.
- e. The computed torque control technique always ensures exponential trajectory tracking. ✔

Your answer is correct.

A corresponding statement that is true is that the computed torque control technique ensures exponential trajectory tracking if the gain matrices K_p and K_v are positive definite and symmetric.

The correct answer is:
The computed torque control technique always ensures exponential trajectory tracking.

Question **3**

Correct

Mark 3.00 out of 3.00

A line passes through the points with coordinates (1,1,1) and (1,-1,1) in a certain axis frame. The coordinates, in the same frame, of five other points are listed in the options below. Four of these points are also located on the line. Indicate which point is **not** located on the line.

- a. (1,3,1)
- b. (1,-5,1)
- c. (1,-10,1)
- d. (1,5,1)
- e. (1,1,3) ✔

Your answer is correct.

The line is clearly parallel to the y axis of the frame. If the position vector $\vec{r}$ of any one of the five points in the options, relative to either one of the two points known to be located on the line, is calculated, and this vector $\vec{r}$ is parallel to the y axis, the point in question lies on the line. Mathematically this is expressed as that the point in question lies on the line if $\vec{j} \times \vec{r}$ is a zero vector.

The correct answer is:
(1,1,3)

Question 4

Correct

Mark 3.00 out of 3.00

Indicate which statement is false.

- a. The trajectory of a rigid body as it moves in 3D space is described by $g_{sb}(t) \in SE(3)$, in terms of a spatial frame S. If, at a certain instant in time, the body velocity $V^b \in \mathbb{R}^6$ of the body is known, the spatial velocity $V^s \in \mathbb{R}^6$ of the body can be calculated by a simple matrix product with one of the factors the adjoint transformation associated with g_{sb}, and the other factor V^b.
- b. The trajectory of a rigid body as it moves in 3D space is described by $g_{sb}(t) \in SE(3)$, in terms of a spatial frame S. Both the body velocity $V^b \in \mathbb{R}^6$ and the spatial velocity $V^s \in \mathbb{R}^6$ of the body can be calculated by calculating the time derivative of $g_{sb}(t)$ and the transpose of $g_{sb}(t)$, and combining them in a matrix product in some way. ✔
- c. The trajectory of a rigid body as it moves in 3D space is described by $g_{sb}(t) = (p_{sb}(t), R_{sb}(t)) \in SE(3)$, in terms of a spatial frame S. If the adjoint transformation associated with g_{sb} is defined as
$$\begin{bmatrix} R_{sb} & \hat{p}_{sb}R_{sb} \\ 0 & R_{sb} \end{bmatrix},$$
then its inverse is
$$\begin{bmatrix} R_{sb}^T & -R_{sb}^T\hat{p}_{sb} \\ 0 & R_{sb}^T \end{bmatrix}.$$
- d. If $\hat{\xi} \in se(3)$ is a twist with twist coordinates $\xi \in \mathbb{R}^6$, then, for any $g \in SE(3)$, $g\hat{\xi}g^{-1}$ is a twist with twist coordinates $\text{Ad}_g\xi \in \mathbb{R}^6$.

Your answer is correct.

A corresponding statement that is true is that the trajectory of a rigid body as it moves in 3D space is described by $g_{sb}(t) \in SE(3)$, in terms of a spatial frame S. Both the body velocity $V^b \in \mathbb{R}^6$ and the spatial velocity $V^s \in \mathbb{R}^6$ of the body can be calculated by calculating the time derivative of $g_{sb}(t)$ and the **inverse** of $g_{sb}(t)$, and combining them in a matrix product in some way.

The correct answer is:
The trajectory of a rigid body as it moves in 3D space is described by $g_{sb}(t) \in SE(3)$, in terms of a spatial frame S. Both the body velocity $V^b \in \mathbb{R}^6$ and the spatial velocity $V^s \in \mathbb{R}^6$ of the body can be calculated by calculating the time derivative of $g_{sb}(t)$ and the transpose of $g_{sb}(t)$, and combining them in a matrix product in some way.

Question 5

Correct

Mark 3.00 out of 3.00

Indicate which statement is false.

- a. The choice of the reference configuration of an open chain manipulator is arbitrary, but the twist coordinates of the individual joints depend on this choice.
- b. The configuration of the tool frame of an open chain manipulator can be calculated at any time t as

  $g_{st}(\theta) = e^{\hat{\xi}_1\theta_1} e^{\hat{\xi}_2\theta_2} \cdots e^{\hat{\xi}_n\theta_n} g_{st}(0)$,

  where $g_{st}(0)$ is the reference configuration of the tool frame.
- c. The configuration of the tool frame of an open chain manipulator can be calculated at any time t as

  $g_{st}(\theta) = g_{l_n t} g_{l_{n-1} l_n}(\theta_n) \cdots g_{l_1 l_2}(\theta_2) g_{s l_1}(\theta_1)$,

  where $g_{l_{i-1} l_i}(\theta_i)$ is the transformation between the adjacent link frames, and $\theta(t) \in \mathbb{R}^n$ is the trajectory of joint angles in the joint angle space.
- d. When using the product of exponentials formula, all vectors and points are specified relative to the base coordinate frame.
- e. The configuration of the tool frame of an open chain manipulator can be calculated at any time t as

  $g_{st}(\theta) = e^{\hat{\xi}_1\theta_1} e^{\hat{\xi}_2\theta_2} \cdots e^{\hat{\xi}_n\theta_n}$,

  on condition that the base frame is chosen to be located and orientated in a very specific way.

Your answer is correct.

A corresponding statement that is true is: the configuration of the tool frame of an open chain manipulator can be calculated at any time t as

$g_{st}(\theta) = g_{s l_1}(\theta_1) g_{l_1 l_2}(\theta_2) \cdots g_{l_{n-1} l_n}(\theta_n) g_{l_n t}$,

The correct answer is:
The configuration of the tool frame of an open chain manipulator can be calculated at any time t as

$g_{st}(\theta) = g_{l_n t} g_{l_{n-1} l_n}(\theta_n) \cdots g_{l_1 l_2}(\theta_2) g_{s l_1}(\theta_1)$,

where $g_{l_{i-1} l_i}(\theta_i)$ is the transformation between the adjacent link frames, and $\theta(t) \in \mathbb{R}^n$ is the trajectory of joint angles in the joint angle space.

◄ Exam B: various files

Jump to...

Exam A question 1 (to be done using the Safe Exam Browser) ►